

Inter-VLAN Routing with DHCP Relay using Cisco Packet Tracer

Project Overview

This project demonstrates enterprise networking concepts using Cisco Packet Tracer. A Cisco 2911 router performs Router-on-a-Stick inter-VLAN routing while a Cisco 2960 switch provides VLAN segmentation. A centralized DHCP server assigns IP addresses using DHCP Relay (ip helper-address).

Objectives

- Configure VLANs
- Configure IEEE 802.1Q trunking
- Implement Router-on-a-Stick
- Configure DHCP Relay
- Verify end-to-end connectivity

Network Components

- Cisco 2911 Router
- Cisco 2960 Switch
- DHCP Server
- PCs
- Ethernet Links

Technologies Used

IEEE 802.1Q, VLANs, IPv4, DHCP Relay, Router-on-a-Stick, Cisco Packet Tracer.

Implementation Summary

Three VLANs were configured. The switch trunk carries VLAN traffic to the router. Router subinterfaces provide gateway functionality for each VLAN. DHCP requests are forwarded to the centralized DHCP server using ip helper-address.

Verification

The following checks confirmed successful implementation:

- show vlan brief
- show interfaces trunk
- show ip route
- show arp
- Successful ping between different VLANs.

Learning Outcomes

Understanding VLAN segmentation, trunking, inter-VLAN routing, DHCP Relay, routing verification, and enterprise network troubleshooting.

Conclusion

The project successfully demonstrates inter-VLAN communication with centralized DHCP services and provides a practical foundation for CCNA and network support roles.